

Flood Risk in Europe

Executive Report — Economic Losses, Human Impact and Climate Risk (EM-DAT, 2000–2026)

*This document summarises, in non-technical language, the findings of the full analysis. The main interactive deliverable is the **Streamlit dashboard**, where every figure below can be explored and filtered live; the full technical analysis behind it, including code, methodology and validation of every figure, is in [analysis.ipynb](#). It is intended for recruiters, hiring managers, or anyone who wants to understand the project and its conclusions without reading code.*

Last updated: July 2026 · Data: EM-DAT (CRED, UCLouvain) · World Bank GDP · Natural Earth

493

flood events analysed

36

European countries

2000–2026

time period

\$93.8B

Germany, total losses

78%

of German losses
uninsured

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In a nutshell

The project analyses 493 flood events recorded across Europe between 2000 and 2026, in 36 countries, using the EM-DAT international disaster database. The goal is to understand where flood risk concentrates, how it has evolved over time, and what it means for insurers, governments and investors who need to assess physical climate risk.

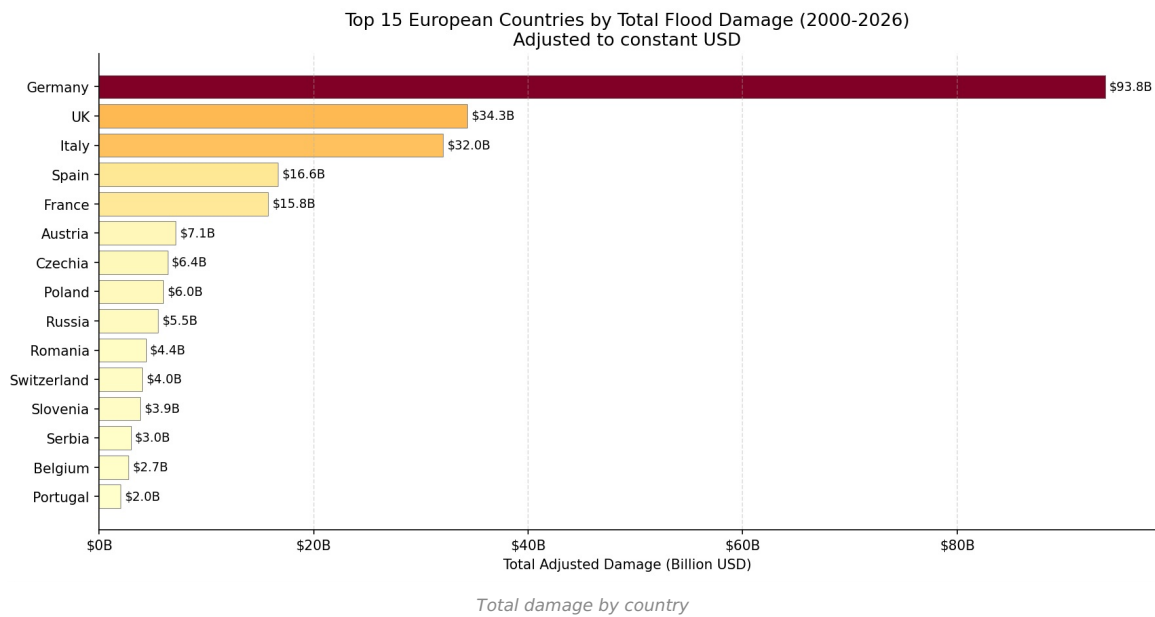
Germany accounts for roughly 38% of Europe's total economic flood losses, driven almost entirely by a single event: the 2021 Ahr Valley floods, which caused 47.5 billion USD in damage. Despite that economic scale, the insurance protection gap remains enormous even in mature markets: 78% of Germany's flood losses were uninsured. In Spain, the human impact has been more severe than the economic one, with 321 cumulative deaths, a large share of them concentrated in a single recent event: the October 2024 DANA storm, which killed 232 people.

Context and objective

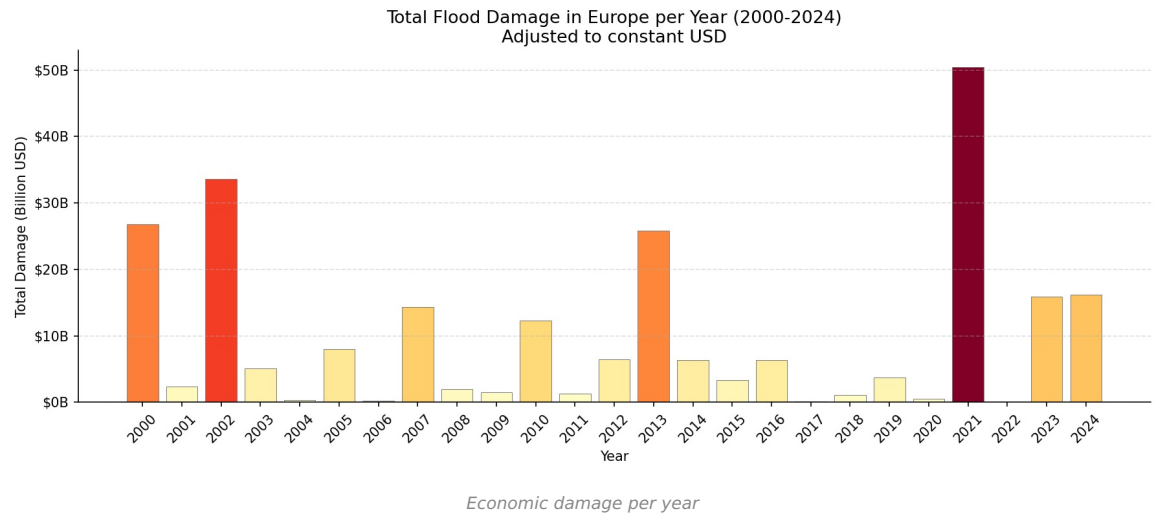
Floods are the most costly and frequent natural disaster in Europe. This analysis set out to answer five business questions: which countries suffer the highest losses, how flood impacts have evolved over time, where the largest insurance gaps are, which flood types are most damaging, and which indicators matter most for physical climate risk assessment. The results are aimed at three audiences with different needs: insurers and reinsurers who need to prioritise risk capital allocation,

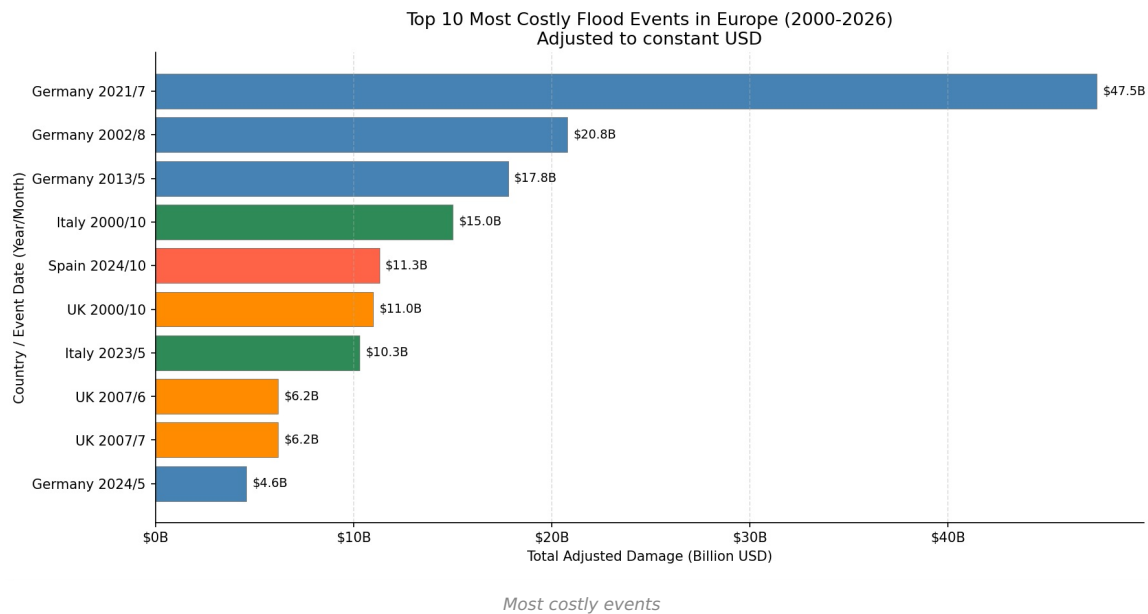
governments and regulators designing adaptation policy, and investors who must assess the physical climate risk of their portfolios under frameworks such as TCFD (Task Force on Climate-related Financial Disclosures).

Economic losses: a striking concentration



Germany dominates cumulative losses by a wide margin, well ahead of the UK and Italy in second and third place. Together, these three countries account for around 66% of all recorded economic damage across the 27 European countries with available data. This pattern is characteristic of catastrophe risk: aggregate losses tend to be driven by a handful of extreme, low-frequency events rather than by the accumulation of smaller ones.

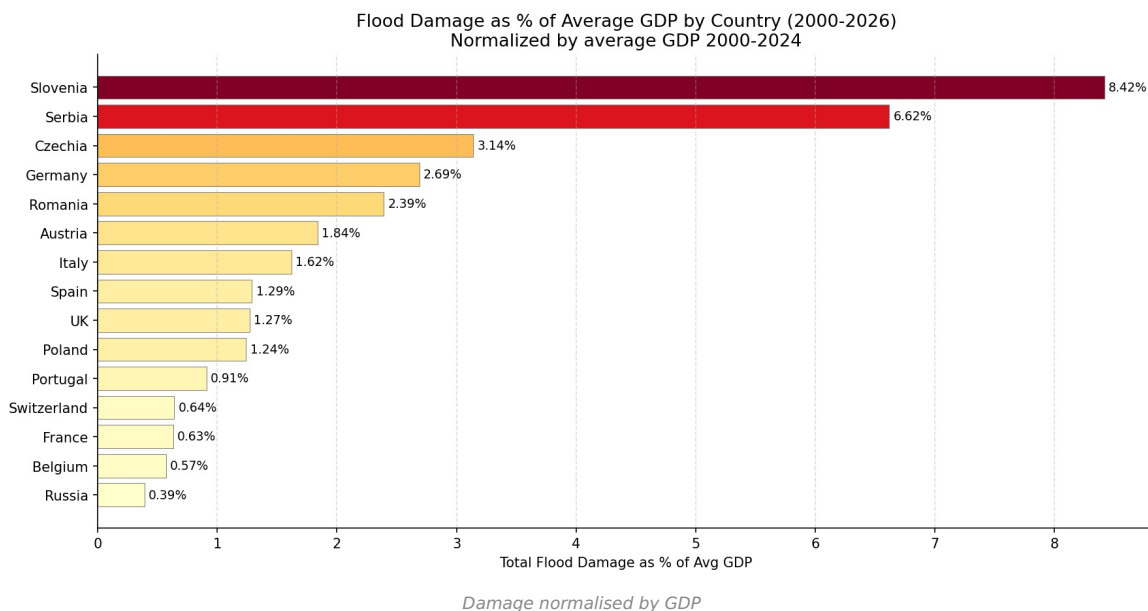




That concentration is clear when looking at individual events: the 2021 Ahr Valley floods alone represent close to 20% of all adjusted flood damage recorded in Europe between 2000 and 2024. Other severe events - Germany in 2002, Italy in 2000, Spain in 2024 - complete a list where a small number of catastrophes explain most of the continent's total losses.

Concentration risk: a single event — the 2021 Ahr Valley floods in Germany (\$47.5B) — accounts for close to 20% of all recorded flood damage in Europe across 26 years. Catastrophe risk is driven by the tail, not the average.

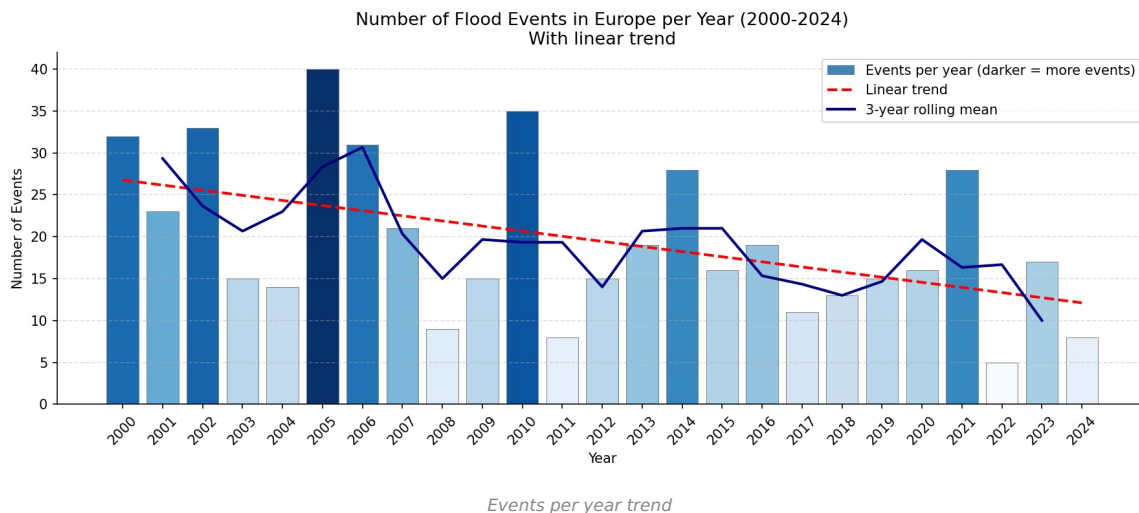
The size of the economy matters



When losses are normalised by the size of each country's economy, the picture changes dramatically. Slovenia and Serbia, which looked minor in absolute terms, emerge as the most exposed countries relative to GDP (8.4% and 6.6% respectively). Germany, despite having the highest absolute losses, drops to fourth place in this relative view, reflecting the much larger size of its economy. This perspective is especially relevant for assessing sovereign fiscal risk and insurance market depth in smaller economies.

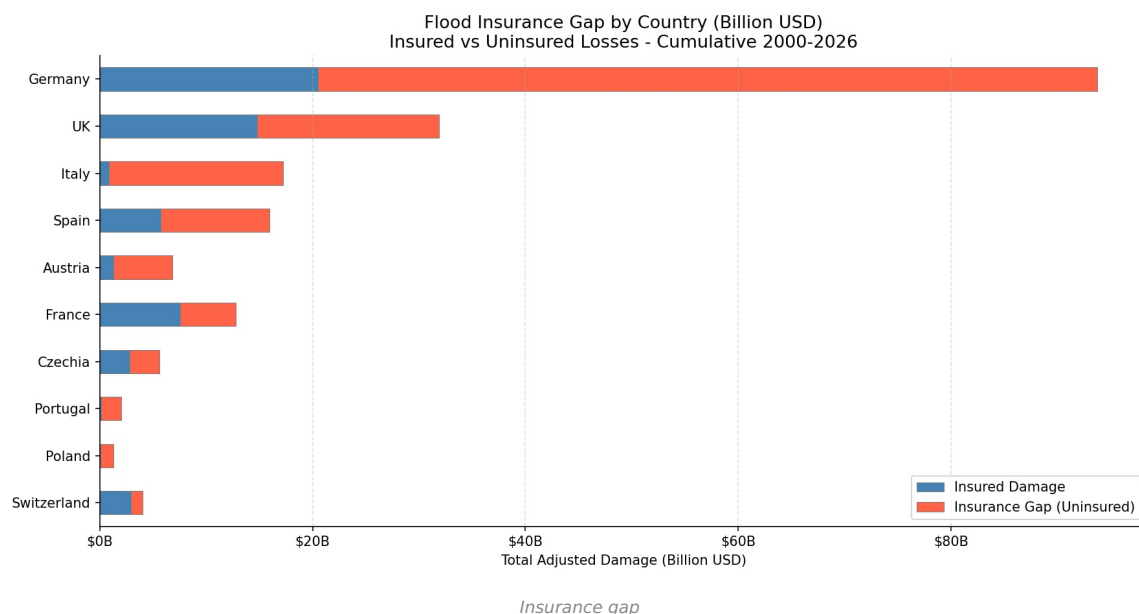
Size matters: Slovenia (8.4% of GDP) is Europe's most exposed country once damage is normalised by economy size — despite ranking outside the top 10 in absolute losses. Absolute and relative risk tell different stories.

Trend: more events, or more severe ones?



The linear trend suggests a slight decline in the number of recorded events per year, but this figure should be read with caution: recent years tend to be underrepresented in disaster databases, since event validation takes time. What is clear is that economic severity is increasing: the 2021 damage spike has no comparison anywhere else in the period analysed. This suggests that while event frequency is not rising dramatically, the magnitude of individual events is - a pattern consistent with IPCC projections on the intensification of extreme climate events.

The insurance protection gap

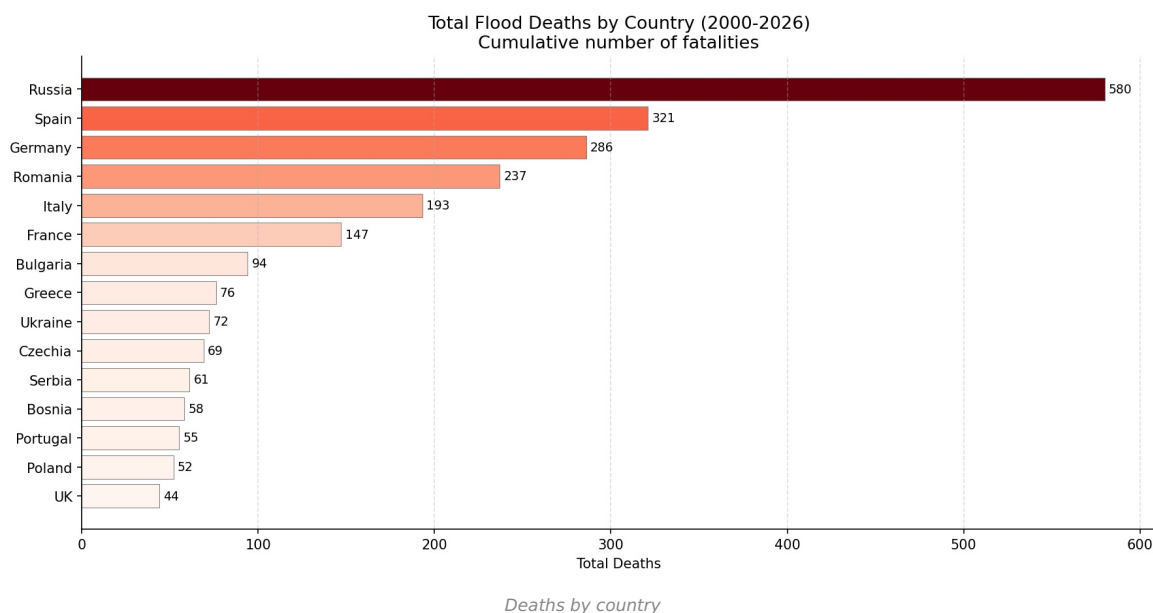


One of the most relevant findings for the insurance sector is the size of the flood protection gap across Europe. In Germany, despite having one of the most developed insurance markets on the continent, 78% of flood losses were uninsured (73.3 billion USD uncovered). In Italy the situation is even more extreme: only 5% of recorded damage was insured. Romania records almost no insured damage data at all, though this may reflect data gaps rather than a genuine absence of coverage.

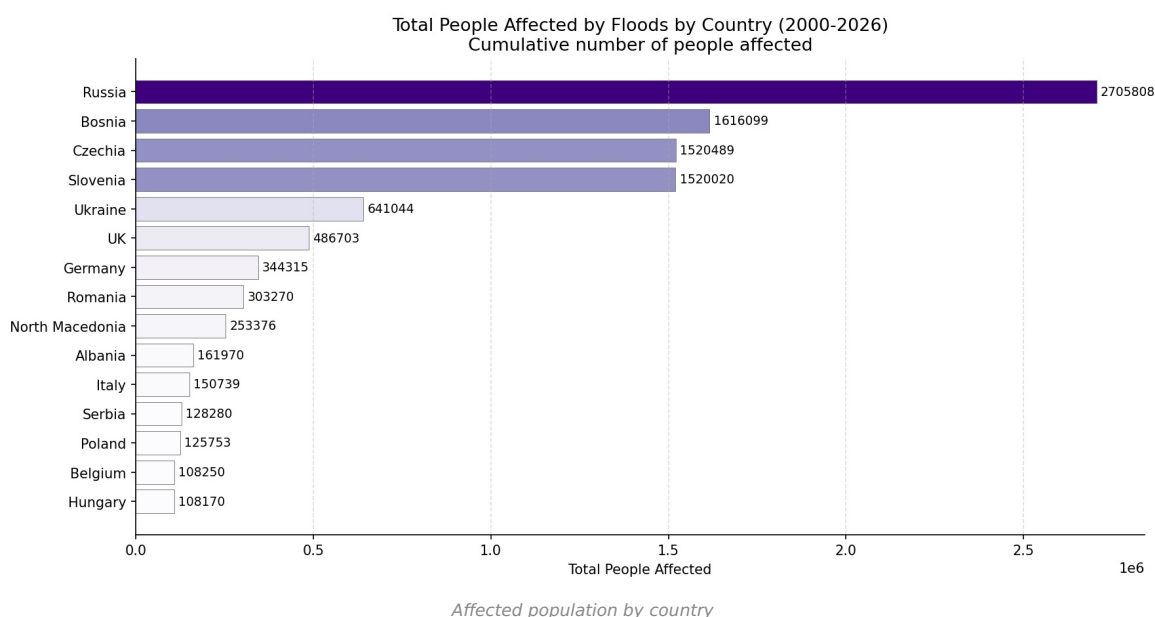
Taken together, these figures support the case for mandatory flood insurance schemes in several European countries.

Protection gap: even Germany's mature insurance market left 78% of flood losses uninsured (73.3B USD). This is not a poorer-country problem — it is systemic across Europe.

Human impact: mortality and affected population

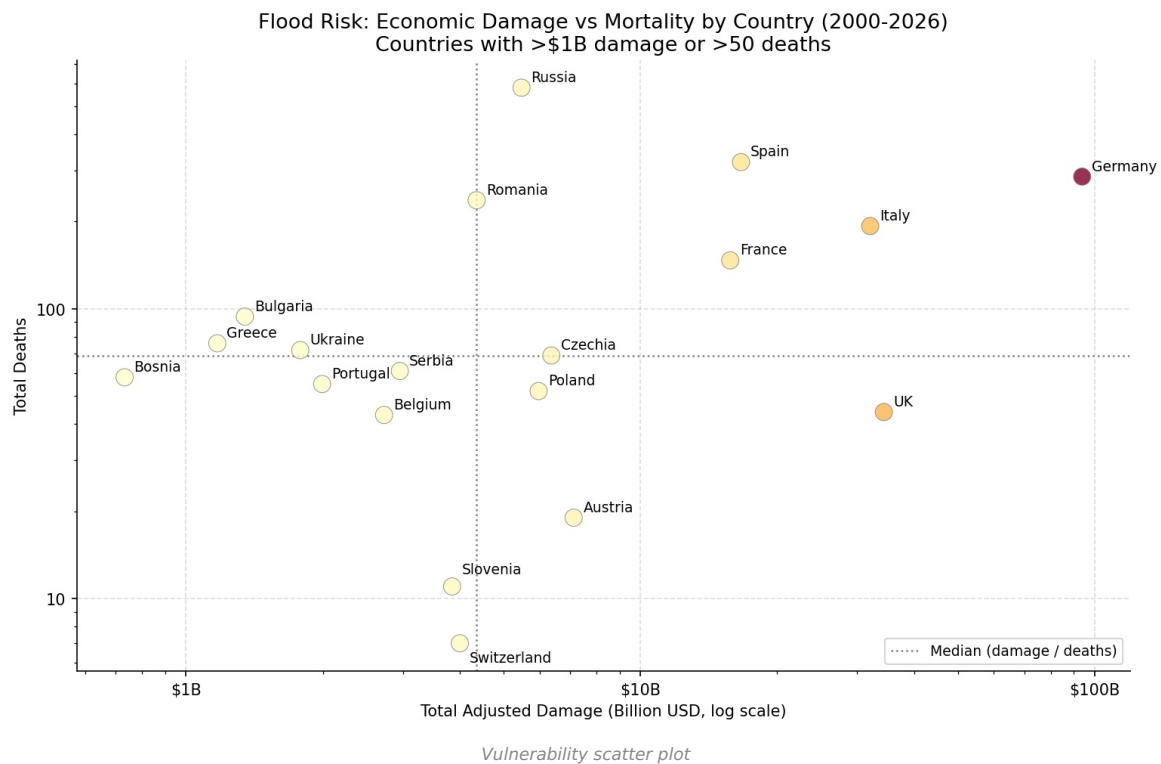


Russia and Spain record the highest cumulative flood mortality, despite suffering considerably lower economic losses than Germany or the UK. This inverse relationship between economic damage and mortality points to differences in resilience, early warning systems and emergency response capacity. In Spain, a large share of deaths are linked to flash floods, including the devastating 2024 DANA event. Eastern European countries such as Romania and Bulgaria also show disproportionately high mortality relative to their economic losses.



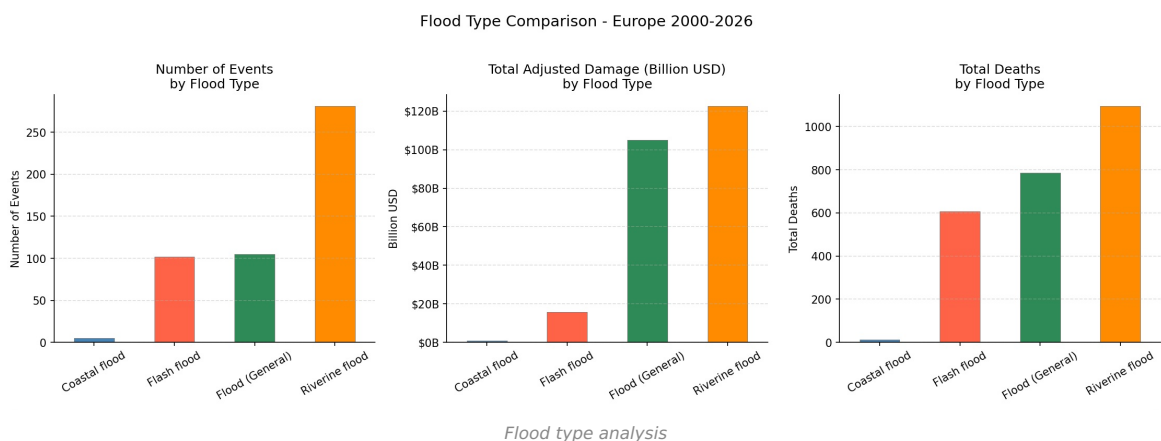
Beyond deaths, the number of people affected tells another part of the story. Russia leads with 2.7 million people affected, consistent with its population size and geography, but Bosnia stands out with 1.6 million affected despite not ranking among the countries with the highest economic damage. Czechia and Slovenia also rank far higher on this measure than in the economic damage tables, showing that human impact and economic impact do not always coincide.

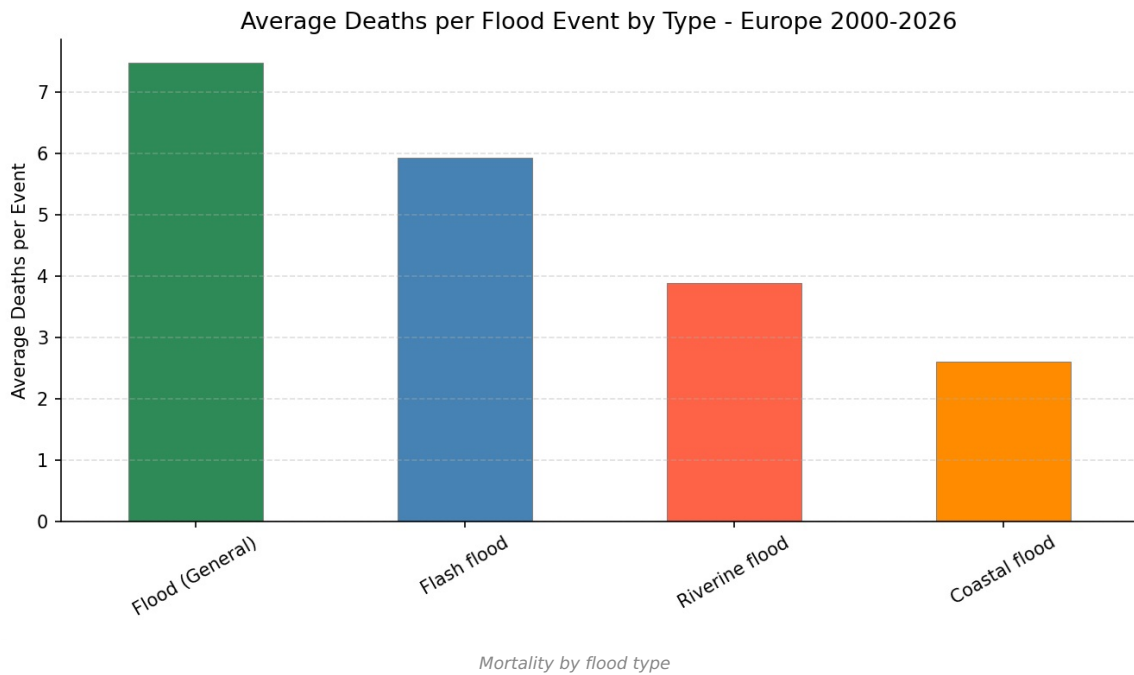
Vulnerability profiles: economic damage doesn't tell the whole story



Cross-referencing economic damage and mortality reveals clearly distinct risk profiles across countries. Romania, Bulgaria, Greece and Ukraine show high mortality despite relatively low economic damage, suggesting lower resilience and adaptive capacity. At the opposite end, the UK, Austria, Czechia and Poland combine high economic losses with comparatively low mortality, reflecting stronger emergency response systems. Germany, Italy, Spain and France fall into the highest-risk quadrant: high economic damage and high mortality at the same time - in fact, this “dual vulnerability” profile turns out to be the most common one among Europe’s larger economies in this dataset, not an exception.

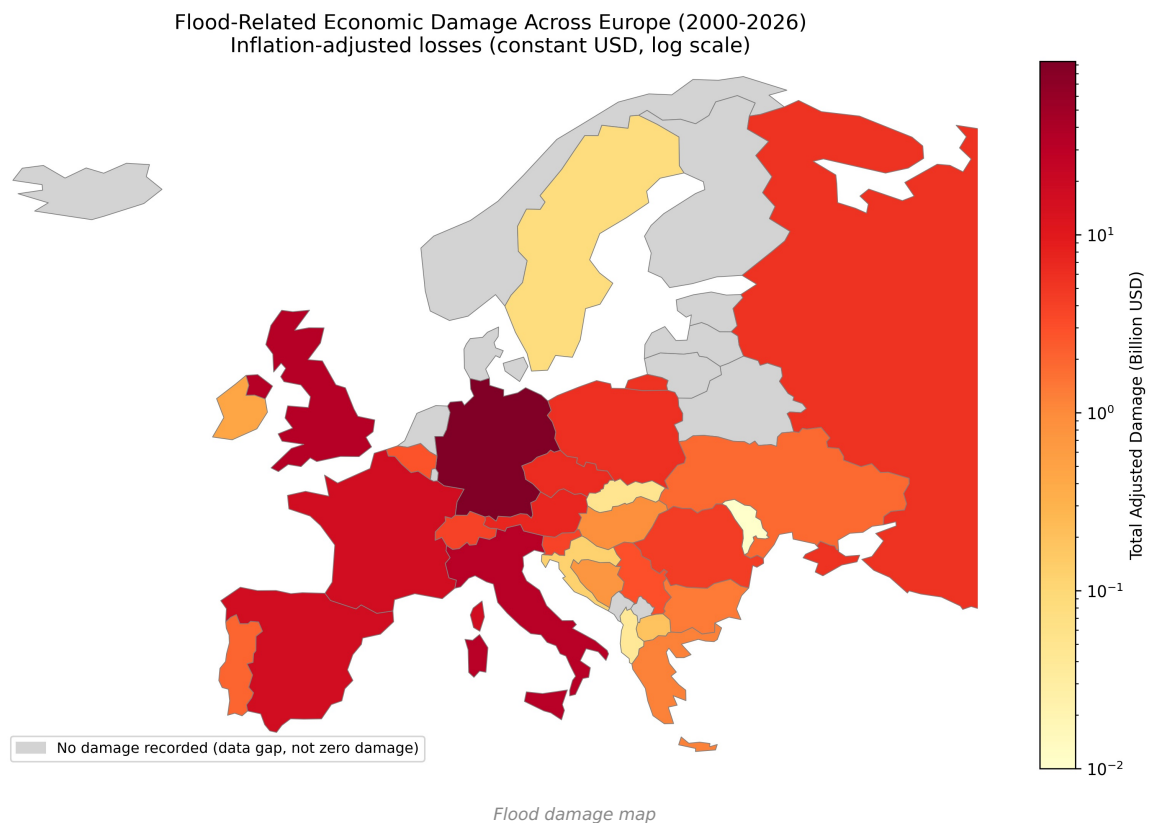
Not all floods are the same





Riverine floods dominate both in frequency (281 events) and cumulative economic damage (122.6 billion USD), but flash floods stand out for their disproportionate lethality: with far fewer events and lower total damage, they account for a significant share of total deaths. This distinction matters for risk management: riverine floods are relatively predictable and slow-developing, allowing time to evacuate; flash floods develop with very little warning, and are projected to become more frequent in Southern Europe under climate change scenarios - Spain's 2024 DANA event is a recent illustration of this lethality.

Where risk concentrates on the map



The map confirms the geographic concentration of losses: Western and Central Europe - particularly Germany, the UK, Italy

and France - account for most of the recorded economic damage, reflecting both greater asset exposure and a higher frequency of severe events. Eastern European countries such as Ukraine show lower recorded losses, which may reflect underreporting rather than genuinely lower exposure. Belarus and Finland have no recorded damage data at all, which reflects a data gap rather than confirmed absence of risk.

What this means for decision-making

For insurers and reinsurers, Germany, Italy and the UK are natural candidates for prioritising risk capital allocation, though always interpreting these figures alongside their respective data coverage rates. Flash floods require separate modelling frameworks given their high mortality rate per event. For governments and regulators, the scale of the insurance gap across Europe supports the case for mandatory flood insurance schemes, while the higher relative mortality observed in Eastern Europe points to a need for targeted adaptation investment. For investors and asset managers, the key takeaway is that flood risk is increasingly concentrated in a small number of catastrophic events, making tail risk assessment critical, and that climate change is expected to intensify both flash floods and coastal floods.

These findings can be organised under the Hazard-Exposure-Vulnerability framework widely used in climate risk assessment: hazard is the flood event itself, differentiated by subtype; exposure is the economic assets located in flood-prone areas; and vulnerability is the capacity - or lack thereof - to withstand and recover from impact, including insurance coverage. Applying this framework to the data: riverine floods combine very high frequency with high economic exposure but medium vulnerability; flash floods combine medium frequency with very high vulnerability; and coastal floods currently show low levels across all three factors, though this is expected to change as sea levels rise.

A note on the limits of the data

It's important to be honest about what this data can and cannot tell us. Only 37% of events recorded in EM-DAT include economic damage figures, and just 10% include insured damage data - so conclusions about losses and insurance gaps should be read as patterns within the available data, not as the complete picture of real-world risk. Recent years tend to be underrepresented due to the natural lag in validating events after a disaster. No historical disaster database can predict future events, but it does provide valuable evidence for identifying vulnerable regions and prioritising physical climate risk assessment.

Conclusion

Flood risk in Europe is not evenly distributed, either economically or in human terms: a small number of extreme events concentrate a disproportionate share of total damage, the insurance protection gap remains systemic even in mature markets, and human impact does not always align with economic impact. Understanding these distinctions - between frequency and severity, between economic damage and mortality, between absolute exposure and exposure relative to economic size - is essential for any serious physical climate risk assessment.

This report summarises the project's main findings. To explore the figures interactively — filtering by country, flood type or year — see the live Streamlit dashboard. The full technical analysis, including code, detailed methodology and validation of every figure, is available in [analysis.ipynb](#).

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GitHub: github.com/AlonsoSheila/flood-risk-europe

Notebook: [analysis.ipynb](#)

Live dashboard: flood-risk-europe.streamlit.app

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